

BRESCIA MONTICHIARI, ItalyWMO: **162593**Lat: **45.429N**Lon: **10.331E**Elev: **109**StdP: **100.03**Time Zone: **1.00 (EUC)**Period: **04-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-4.2	-2.9	-11.2	1.5	3.9	-8.4	1.9	2.8	9.0	9.0	7.5	6.4	2.1	280	0.431

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.8	33.2	21.8	32.0	21.4	30.9	20.8	23.1	30.6	22.5	29.7	21.7	28.7	3.1	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.9	15.7	27.1	20.0	14.9	26.1	19.1	14.1	25.5	68.8	30.6	66.4	29.9	64.0	29.1	26.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.3	6.9	5.8	-7.0	35.4	2.3	1.3	-8.6	36.4	-9.9	37.1	-11.2	37.8	-12.8	38.8		
(5)			-7.2	24.4	2.2	1.2	-8.8	25.2	-10.1	25.9	-11.4	26.5	-13.0	27.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	14.3	3.2	5.0	9.7	14.4	18.3	23.0	25.3	24.4	20.3	14.9	9.0	3.7
	DBStd	8.31	2.79	3.02	3.17	2.98	3.29	3.08	2.42	2.82	2.86	3.06	3.35	3.11
	HDD10.0	650	213	139	43	3	0	0	0	0	0	2	56	194
	HDD18.3	2134	471	372	266	123	42	3	0	1	13	112	279	453
	CDD10.0	2225	1	1	34	133	257	389	473	447	308	154	26	1
	CDD18.3	666	0	0	0	3	41	142	215	189	71	5	0	0
	CDH23.3	6041	0	0	2	34	307	1279	2177	1794	428	18	0	0
	CDH26.7	2152	0	0	0	2	66	448	857	684	93	2	0	0
Wind	WSAvg	2.6	2.3	2.5	2.9	3.0	2.9	2.8	2.8	2.6	2.6	2.3	2.5	2.2
Precipitation	PrecAvg	896	66	59	65	65	84	82	74	87	68	92	93	56
	PrecMax	1264	155	127	159	116	151	181	177	183	209	297	155	95
	PrecMin	617	5	0	1	27	10	9	14	11	5	0	28	20
	PrecStd	181	45	35	45	27	48	47	49	52	61	80	41	24
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.9	16.0	22.2	26.2	30.9	34.0	35.0	35.1	30.9	25.8	18.2	13.0
		MCWB	9.7	9.5	12.3	15.9	18.7	21.5	22.0	21.9	19.9	17.6	14.7	10.1
	2%	DB	10.2	13.0	19.8	24.1	28.2	32.1	33.1	33.8	29.2	23.0	16.2	11.0
		MCWB	7.2	8.4	11.6	15.0	17.9	21.2	22.0	21.5	19.7	16.9	13.5	8.8
	5%	DB	8.9	11.2	17.8	22.2	26.8	30.9	32.0	31.8	27.8	21.8	14.9	9.9
		MCWB	6.4	7.4	11.0	14.1	17.4	20.8	21.6	21.2	19.2	16.4	12.8	8.2
	10%	DB	7.2	9.9	15.8	20.8	24.9	29.2	30.9	30.1	25.9	20.0	13.8	8.2
		MCWB	5.8	6.9	10.1	13.7	16.7	19.8	21.1	20.6	17.9	15.5	12.0	7.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.8	10.3	13.6	16.8	20.3	23.6	24.2	23.9	21.6	19.1	15.8	11.1
		MCDB	12.6	14.3	18.6	23.4	27.3	31.5	31.6	31.1	27.8	22.4	17.2	11.7
	2%	WB	8.2	9.1	12.6	15.8	19.1	22.5	23.2	22.8	20.8	18.1	14.2	9.5
		MCDB	9.2	11.7	17.8	22.1	26.0	30.2	30.8	30.0	26.8	21.4	15.7	10.2
	5%	WB	7.1	8.3	11.6	15.1	18.2	21.5	22.6	22.2	20.0	17.3	13.2	8.2
		MCDB	7.9	10.2	16.1	20.8	24.7	28.5	29.9	29.4	25.5	20.3	14.6	9.2
	10%	WB	6.1	7.5	10.8	14.2	17.3	20.7	21.9	21.5	19.2	16.4	12.4	7.1
		MCDB	6.9	8.9	14.4	19.5	23.5	27.6	28.8	28.4	24.5	19.1	13.7	7.9
Mean Daily Temperature Range	5% DB	MDBR	6.5	7.6	9.2	9.2	9.4	9.5	9.8	9.6	8.9	7.6	6.1	6.5
		MCDBR	9.1	10.7	12.8	12.1	11.2	10.9	10.6	10.8	10.5	9.6	7.5	8.3
		MCWBR	6.4	6.8	6.6	5.3	4.2	3.9	3.7	3.4	3.5	4.4	4.9	6.0
	5% WB	MCDBR	7.1	8.7	10.6	10.9	10.3	10.2	9.6	9.9	9.2	7.7	6.2	6.4
		MCWBR	5.5	5.9	5.9	5.1	4.4	4.2	3.8	3.8	3.5	3.9	4.2	4.9
Clear-Sky Solar Irradiance	taub	0.348	0.378	0.425	0.460	0.458	0.484	0.466	0.454	0.444	0.432	0.378	0.342	
	taud	2.375	2.260	2.135	2.072	2.128	2.103	2.163	2.208	2.201	2.186	2.302	2.401	
	Ebn at Noon	743	784	793	799	816	794	804	798	767	708	691	710	
	Edh at Noon	77	103	133	154	151	155	145	133	124	110	82	68	
All-Sky Solar Radiation	RadAvg	1.42	2.23	3.54	4.50	5.57	6.37	6.53	5.57	4.12	2.48	1.47	1.22	
	RadStd	0.22	0.33	0.47	0.50	0.51	0.35	0.36	0.36	0.28	0.36	0.22	0.16	